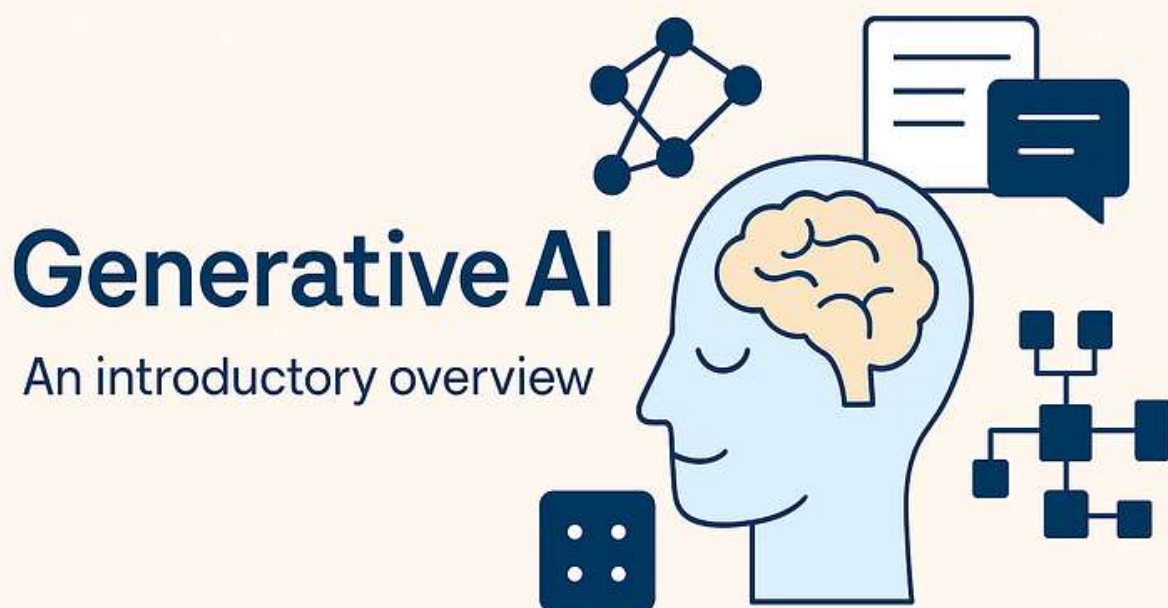

GENAI 101

This can be starter wiki for everyone who are not getting correct path to start Gen AI.

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What We Will Learn Today

By the end, you will clearly understand what Gen AI is, how it works, and how to build real applications using it .

AI Work in IT Companies

Before proceeding further, let's understand **how IT companies are actually working with AI**.

Most IT companies adopt AI tools and technologies in **three major ways**, and accordingly, developers can be broadly classified into **three levels** based on their involvement with AI.

L1 Developers — AI / ML Engineers

These developers work **deeply with AI models** and core AI technologies.

They are involved in:

- Model training and fine-tuning
- Model optimization and cost optimization
- Model deployment and monitoring
- Working with ML algorithms, data pipelines, and infrastructure

✓ Requires strong knowledge of:

- Machine Learning / Deep Learning
- Statistics and algorithms
- Python, PyTorch, TensorFlow, MLOps

L2 Developers — AI Application Developers

These developers **consume AI models** and integrate them into real-world applications.

They are involved in:

- Using APIs of LLMs (OpenAI, Azure OpenAI, Anthropic, etc.)
- Prompt engineering and chaining
- RAG (Retrieval-Augmented Generation)
- Adding AI features to web, mobile, or enterprise applications

✓ Requires skills in:

- Application development (.NET, Java, JS, etc.)
- LLM usage patterns
- System design with AI components

L3 Developers — AI Tool Users

These developers use **AI-powered developer tools** to improve productivity.

Examples include:

- GitHub Copilot

- Cursor AI
- Replit AI
- Factory Droid and similar tools

They mainly:

- Generate code using prompts
- Refactor or debug code using AI
- Speed up routine development tasks

✓ Useful for productivity, but **limited AI depth**

Key Note

If you are currently an **L3 developer**, think of it as a **starting point**, not a risk state.

However, **long-term career growth requires moving toward L2.**

! Developers who *only* rely on AI tools without understanding how AI is applied in systems may find their roles becoming less valuable over time.

✓ The safest and most future-proof path: **Upgrade from L3 → L2**, and optionally toward L1 if interested.

Companies are taking paid subscriptions of GitHub copilot and cursor AI tools and providing assistance to Developers to use it and get work done in few days.

Let's Deep Dive in Gen AI

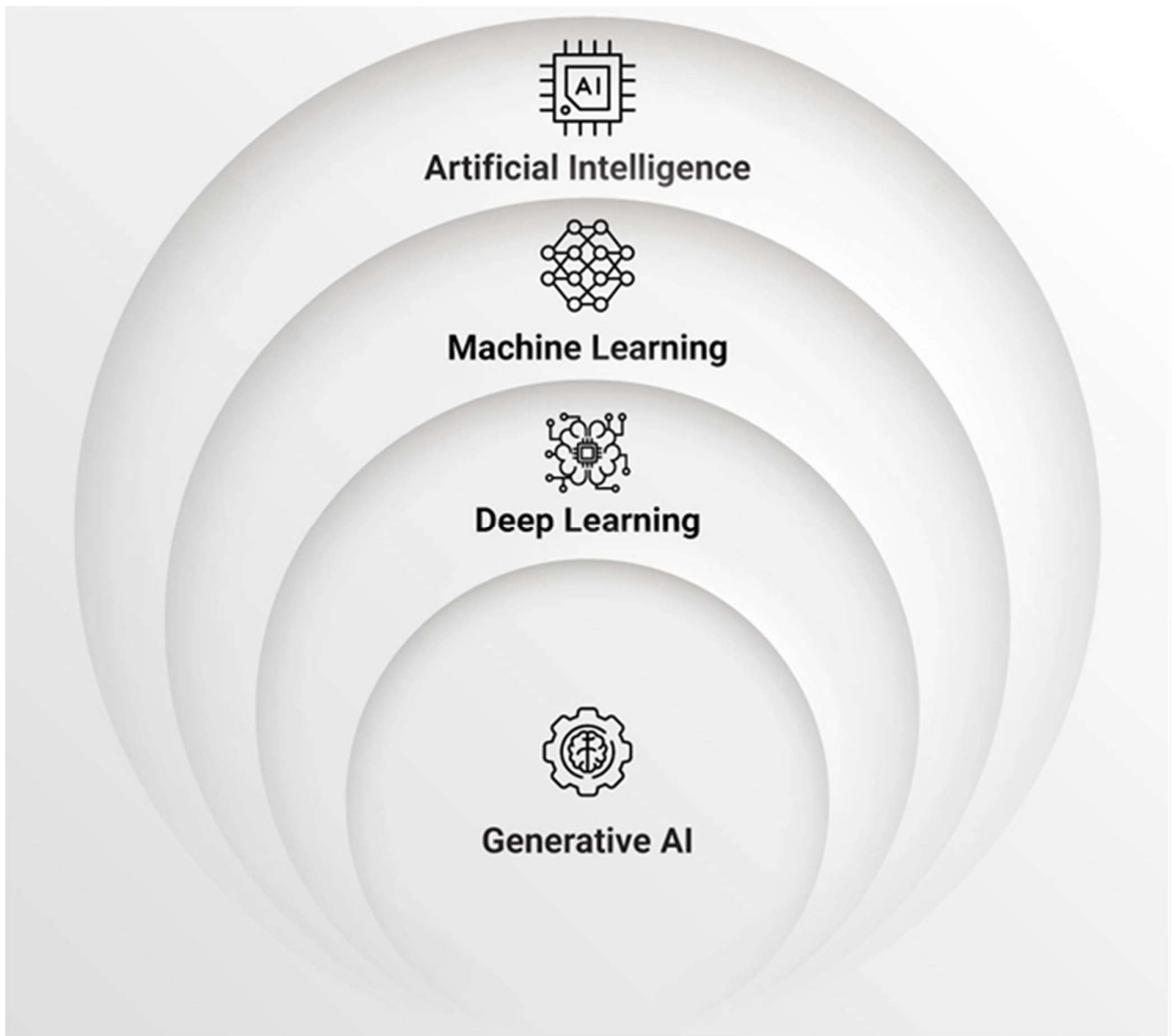
Road Map for becoming GEN AI Developer will be below.

1. Need to Understand Where Gen AI lies in bigger circle of AI
2. What is Difference between Gen AI and Machine Learning / Deep learning
3. Traditional AI vs Generative AI
4. What is Gen AI and how it works.
5. What are Tokens, Embeddings, Vector DB, context window, Context Engineering, Prompt Engineering, Position Encoding, Self-Attention etc
6. Type of Models we have and their Providers.
7. Open-Source Models and Closed Source Models
8. What programming language we need to create GenAI apps.
9. Examples of Gen AI Apps.
10. Understanding Model Parameters
11. Understanding Cost & Billing

Let's Learn in Details

1. Need to Understand Where Gen AI lies in bigger circle of AI

Gen AI is Part of AI and Subset of ML.



Gen AI

```
Artificial Intelligence (AI)
├─ Machine Learning (ML)
│   └─ Deep Learning (DL)
│       └─ Generative AI (Gen AI)
```

a. **Gen AI** is part of AI and Subset of Machine learning.

b. Here we are using pretrained Models or LLM created via Machine learning.

c. **Generative AI** generates new data like Text/ Image / Video/ programming Code via existing Models like GPT 5 , Google Gemini / Claude LLMs

2. What is Difference between Gen AI and Machine Learning / Deep learning

a. Machine Learning

It is a subset of Artificial Intelligence where systems learn patterns from data to make predictions or decisions.

Main focus:

- Prediction
- Classification
- Recommendation

Examples

Spam vs non-spam email
Credit score prediction
Sales forecasting
Recommendation engines (Netflix, Amazon)

b. Deep Learning (DL)

Deep Learning is a subset of Machine Learning that uses multi-layer neural networks to learn complex patterns.

Main focus:

- Feature learning automatically
- Handling large and complex datasets

Examples

Image recognition
Speech-to-text systems
Face recognition
Autonomous driving (perception systems)

c. Gen AI

Generative AI is a subset of Deep Learning that focuses on creating new data/content similar to human-created content.

Here we are using Existing Pre trained Models like GPT , Google Gemini in our applicaitons.

Main Focus

- Content creation
- Creativity & generation

Examples

ChatGPT writing stories & answers
DALL·E generating images
GitHub Copilot generating code
Text summarization
Video generation

3. Traditional AI vs Generative AI

Traditional AI — Predicts or classifies data

Example:

Train a model with labeled fruit/vegetable images

Classify whether new input is a fruit or vegetable

Generative AI — Creates entirely new content

Example:

- Generates a paragraph, image, or summary based on input

4. What is Gen AI and how it works.

Generative AI refers to AI systems that **create new content** instead of merely analyzing or predicting based on existing data.

It can generate

- 📄 Text (articles, code, summaries)
- 🖼️ Images
- 🎵 Audio & music
- 🎬 Videos

Examples

- **ChatGPT → Text generation**
- **DALL·E → Image generation**

We typically **do not train models from scratch**.

Instead, we **consume pre-trained models** such as:

- GPT
- Gemini
- LLaMA

We build **Gen AI applications programmatically**.

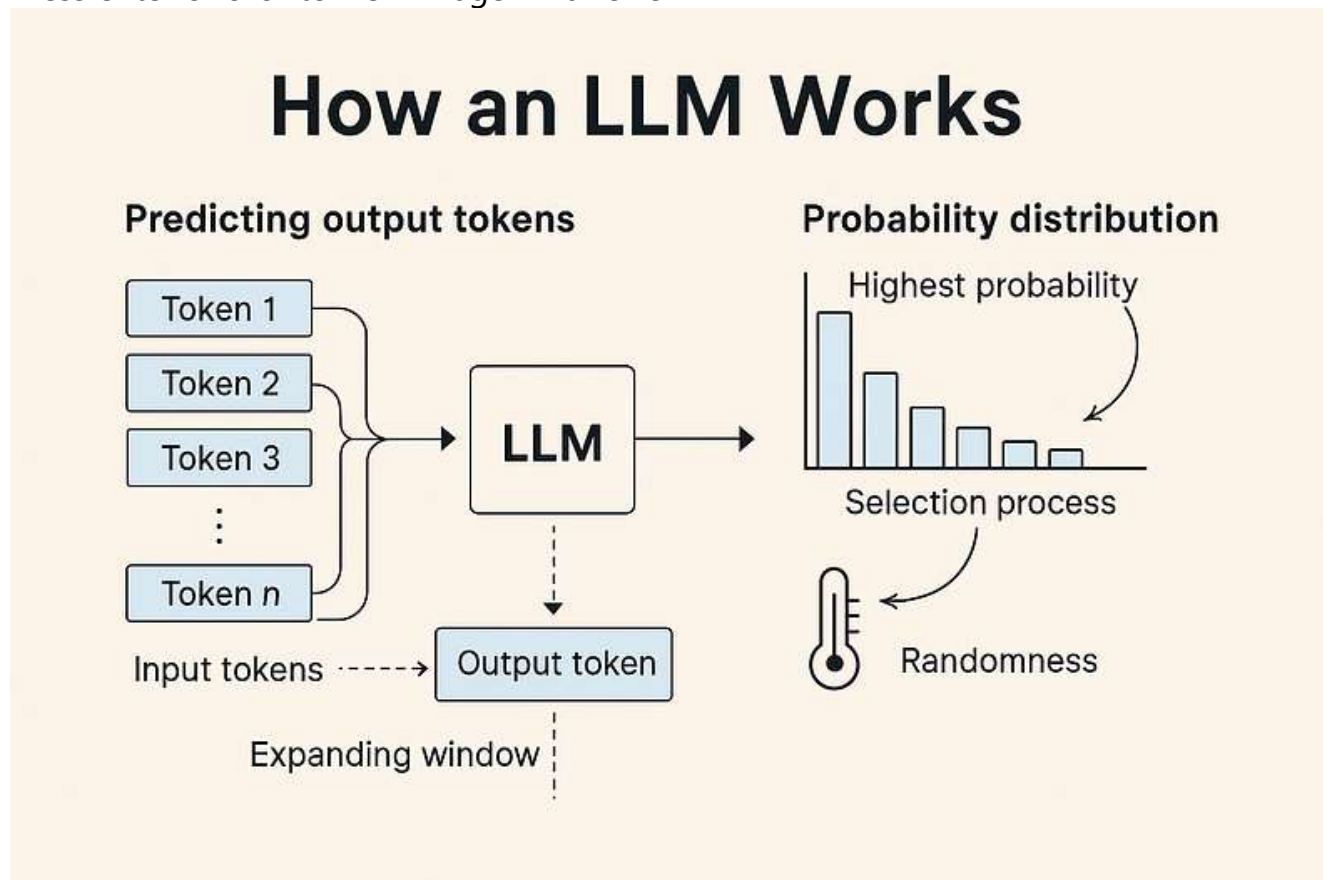
5. How LLM Generate Text.

LLM is capable to generate text via predictions only, it does not have capability to solve mathematical problems.

LLM works on **probability, patterns, and numbers** — not human-like understanding.

LLM has been trained on : Public websites, books, articles, forums, open-source code, and licensed datasets. LLM has trained on huge data, you can say billions of parameters.

Every Time when we ask something it predicts next word.
Press enter or click to view image in full size



Suppose User Input is “What is Color of Rose” -

1. LLM does not understand English, everything what ever we ask — it get converted into numerical value we called as **Vectors or Embeddings**
2. “The color of rose is” will be converted into token first then vectors

3. **Tokens** — [“The”, “color”, “of”, “rose”, “is”]

4. **Tokens** → **Embeddings** (Vector Conversion) token converted into Numerical values called embeddings.

5. Positional Encoding -> Helps the model understand sequence position

6. Self-Attention (Multi-Head Attention) -> Determines which words should focus on other words, In our example “Color of rose is” so self-attention will be on rose and color only rest of words will be ignored for next word calculation

7. Context window- Size of input text, there is limit of input text in LLM

8. **Embeddings** of input value will be like below

[“The”, “color”, “of”, “rose”, “is”] — [1012, 3456, 231, 8743, 112]

6. Probability Calculation (Next Token Prediction)

Now the model predicts the **next token**, Internally, it calculates probabilities:

like below

Word	Probability
red	0.72
pink	0.15
white	0.08

beautiful	0.02
sweet	0.01

LLM assign probability to next possible tokens as above, red has highest probability and sweet has least so LLM choose token or word which has highest probability

6. Type of Models we have and its Providers

Model and LLM are same thing so do not confuse.

- a. GPT-4 / GPT-4.1 — *OpenAI*
- b. Claude 3 — *Anthropic*
- c. Mistral / Mixtral — *Mistral AI*
- d. Gemini 1.5 — *Google*
- e. LLaMA 3 — *Meta*

7. Integration of LLM with Applications

There are multiple ways to do that. All Model providers provide their API's and Keys to connect Api to get data

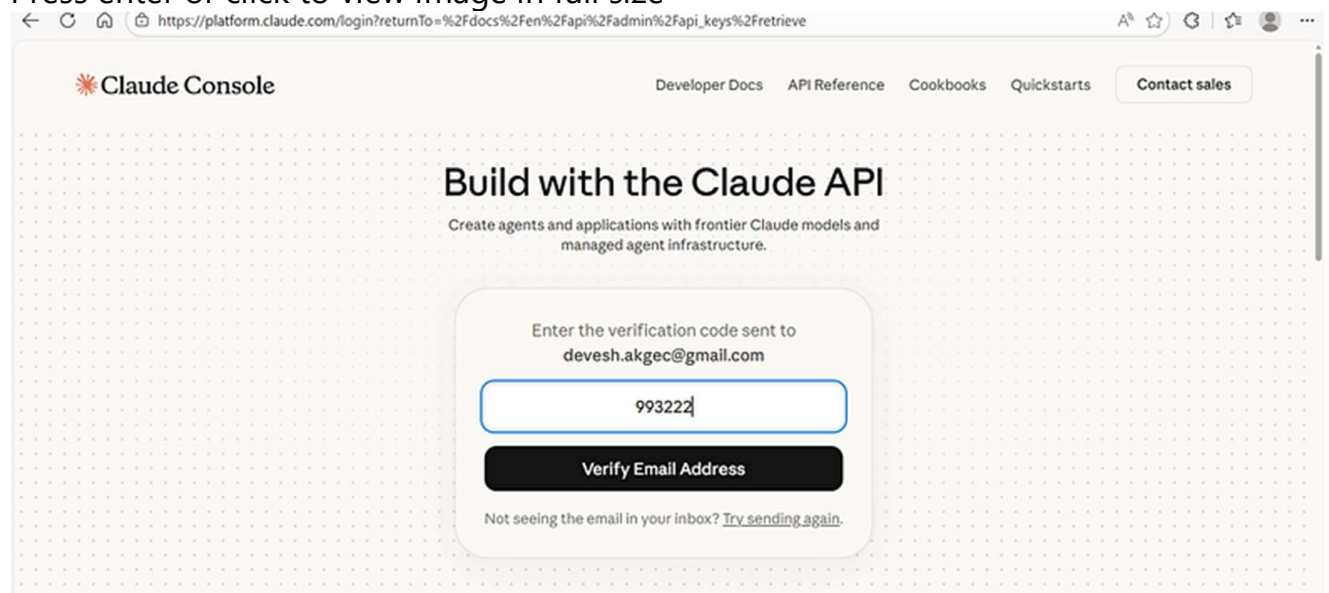
We can easily generate Text/ Video / Image from these API's

We need to pay for keys from Claude / Open AI and Google Gemini but google Gemini provide free subscriptions also and have rate limits.

Steps to get Claude Key

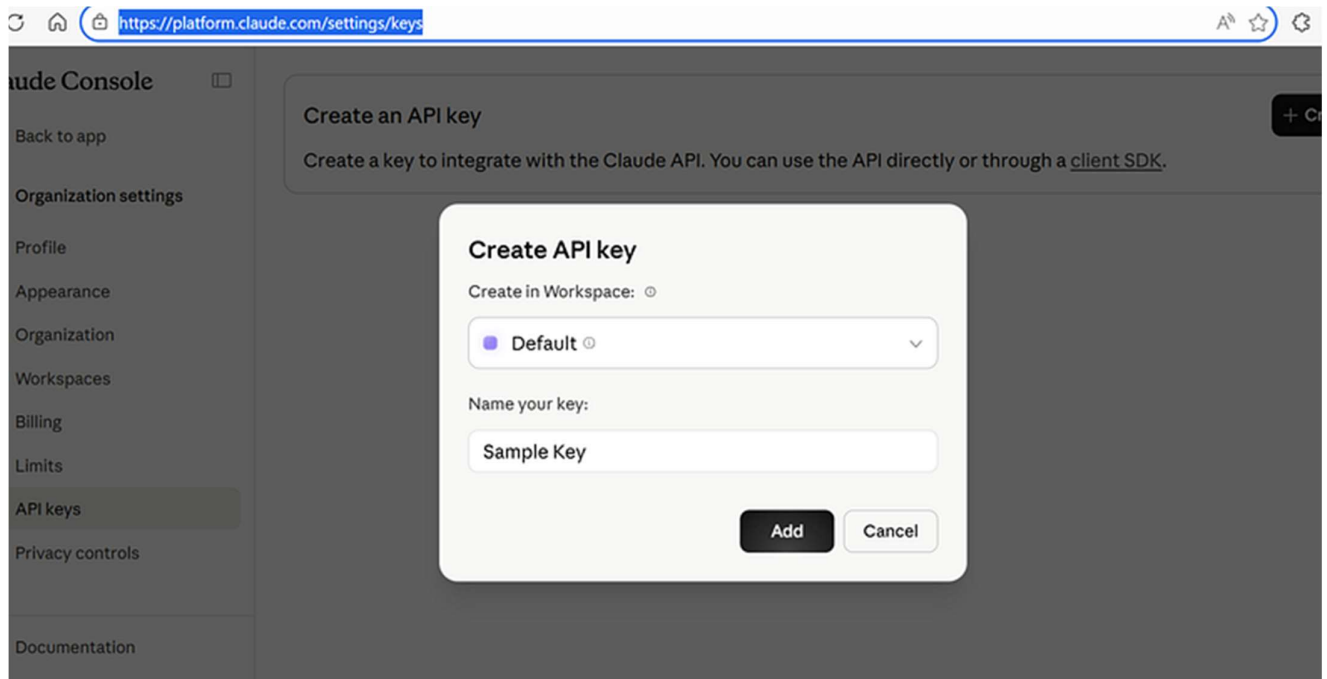
[API Keys | Claude Platform](#)

Press enter or click to view image in full size



Generate Key and Copy.

Press enter or click to view image in full size



Calling API via multiple ways

a. With Python

b. With Curl commands

c. With Typescript

Curl Command to Get Text

```
curl https://api.anthropic.com/v1/messages \
-H "Content-Type: application/json" \
-H "x-api-key: $ANTHROPIC_API_KEY" \
-H "anthropic-version: 2023-06-01" \
-d '{
  "model": "claude-opus-4-6",
  "max_tokens": 1000,
  "messages": [
    {
      "role": "user",
      "content": "What should I search for to find the latest
developments in renewable energy?"
    }
  ]
}
```

```
]
}'
```

Please replace above key which you generated above.

Output will be like this

```
{
  "id": "msg_01HCDu5LRGeP2o7s2xGmxyx8",
  "type": "message",
  "role": "assistant",
  "content": [
    {
      "type": "text",
      "text": "Here are some effective search strategies to find the
latest renewable energy developments:\n\n## Search Terms to Use:\n-
\"renewable energy news 2024\"\n- \"clean energy breakthrough\"\n-
\"solar/wind/battery technology advances\"\n- \"green energy
innovations\"\n- \"climate tech developments\"\n- \"energy storage
solutions\"\n\n## Best Sources to Check:\n\n**News & Industry Sites:**\n-
Renewable Energy World\n- GreenTech Media (now Wood Mackenzie)\n- Energy
Storage News\n- CleanTechnica\n- PV Magazine (for solar)\n- WindPower
Engineering & Development..."
    }
  ],
  "model": "claude-opus-4-6",
  "stop_reason": "end_turn",
  "usage": {
    "input_tokens": 21,
    "output_tokens": 305
  }
}
```

Result will be getting inside content

```
"content": [
  {
    "type": "text",
    "text": "Here are some effective search strategies to find the
latest renewable energy developments:\n\n## Search Terms to Use:\n-
\"renewable energy news 2024\"\n- \"clean energy breakthrough\"\n-
\"solar/wind/battery technology advances\"\n- \"green energy
innovations\"\n- \"climate tech developments\"\n- \"energy storage
solutions\"\n\n## Best Sources to Check:\n\n**News & Industry Sites:**\n-
Renewable Energy World\n- GreenTech Media (now Wood Mackenzie)\n- Energy
```



```
Storage News\n- CleanTechnica\n- PV Magazine (for solar)\n- WindPower  
Engineering & Development..."  
}
```